

Assignment 1

Explore following 4 websites and answer the questions

- bitcoin.org
- Blockchain.com
- <https://coinmarketcap.com/>
- <https://www.investopedia.com/>

One-Word Answerable Questions from
bitcoin.org

Bitcoin Basics

1. Bitcoin was introduced by whom?
Satoshi Nakamoto
2. Bitcoin transactions are recorded on which data structure?
Blockchain
3. Bitcoin uses which consensus mechanism?
Proof-of-Work
4. The smallest unit of Bitcoin is called?
Satoshi
5. Bitcoin is a peer-to-peer type of what system?
Payment

Wallets

6. A wallet stores the user's private key.
7. The recovery phrase used in wallets is called a seed phrase.
8. Hardware wallets store keys offline.
9. Bitcoin addresses are derived from public keys.
10. A private key is required to sign a transaction.

Security

11. Bitcoin prevents double-spending using the blockchain.

12. Bitcoin is secured using asymmetric cryptography.
13. To run a Bitcoin node, the software required is Bitcoin Core.
14. The process of verifying transactions is called validation.
15. The Bitcoin network is decentralized.

Mining

16. Miners solve cryptographic puzzles to add blocks.
17. Mining rewards a block with newly created Bitcoins.
18. The block time target for Bitcoin is 10 minutes.
19. Mining difficulty adjusts every 2016 blocks.
20. Miners compete to find a valid block hash.

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Blockchain Structure

21. Each block contains a Merkle root.
22. Blocks are linked using the previous block's hash.
23. A Bitcoin transaction contains inputs and outputs.
24. Unspent transaction outputs are called UTXOs (Unspent Transaction Outputs).
25. The first-ever Bitcoin block is called the Genesis block.

Network

26. A full node stores the entire blockchain ledger.
27. Bitcoin uses the P2P (Peer-to-Peer) protocol for communication.
28. A light wallet uses SPV mode.
29. Nodes relay information across a peer-to-peer network.

A full node is a program that fully validates transactions and blocks. Almost all full nodes also help the network by accepting transactions and blocks from other full nodes, validating those transactions and blocks, and then relaying them to further full nodes.

List any two developer resources available on bitcoin.org

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Wallets and Payment Processing

Explore : <https://coinmarketcap.com/>

Task C1: Market Information

- Current price of BTC: \$64,020.76
- 24-hour % change: +0.58%
- Market capitalization: \$1.28T
- Circulating supply: 20.06M BTC
- Total/maximum supply: 20.06M BTC
- Task C2: Historical Data
- What was the price of Bitcoin one year ago?
\$115,072.18
- Note the highest and lowest prices in the last 30 days.
Lowest \$61,793.49 , Highest \$ 66,855.57

Explore <https://www.investopedia.com/>

Why is Bitcoin considered volatile?

Bitcoin is considered volatile because its price can change rapidly due to market demand and supply, investor sentiment, regulatory news, macroeconomic events, and its relatively limited supply.

List two investment risks mentioned.

Market volatility (price fluctuations)

Regulatory uncertainty (changes in government laws and regulations)

Part E: Reflection & Analysis

Which website gave you the most clear understanding? Why?

Bitcoin.org gave me the clearest understanding because it explains Bitcoin, blockchain, wallets, mining, and security in a simple and beginner-friendly manner. The information is well organized and focuses on the fundamentals, making it easier to understand how Bitcoin works.

In one paragraph, explain how blockchain ensures trust without a central authority.

Blockchain ensures trust without a central authority by storing transactions on a distributed ledger that is shared among thousands of computers (nodes). Every transaction is verified using a consensus mechanism before being added to a block. Each block is linked to the previous one using cryptographic hashes, making the data extremely difficult to alter. Since all participants maintain copies of the blockchain and independently verify transactions, no single person or organization controls the network, ensuring transparency, security, and trust.